



Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY TECHNICAL BOOTCAMP · 40 HOURS

AI on Google Cloud

Vertex AI, Gemini, Document AI, Dialogflow, and the complete GCP AI stack

5

DAYS

40

HOURS

20+

HANDS-ON LABS

6

GCP PROJECTS

Course Overview

This 5-day hands-on bootcamp covers the full Google Cloud AI and ML platform — from Gemini API and Vertex AI through Document AI, Dialogflow CX, AutoML, and BigQuery ML. Developers build 6 production-ready AI solutions using Google's AI stack, learning to leverage its strengths in multimodal AI, data analytics integration, and open-source model hosting.

Google's AI stack is uniquely powerful for data-rich organizations. Gemini's multimodal capabilities, Vertex AI's integrated ML platform, BigQuery ML's SQL-based modeling, and Dialogflow's conversational AI — this bootcamp covers the entire ecosystem hands-on. You'll understand how Google Cloud AI competes with and complements Azure and AWS, and when GCP is the right choice for your workload.

Who Is This For?



GCP Developers

Engineers building AI solutions on Google Cloud



Data Engineers

BigQuery users extending into ML and AI



ML Engineers

Engineers deploying models on Vertex AI



Multi-Cloud Teams

Teams evaluating GCP AI alongside Azure/AWS

Learning Outcomes

- 1 **Build with Gemini API** — multimodal prompting (text, image, video, audio), function calling, grounding, system instructions, and streaming
- 2 **Master Vertex AI platform** — model deployment, Vertex AI Pipelines, model registry, endpoints, evaluation, and experiment tracking
- 3 **Build with AutoML & BigQuery ML** — train custom models without code using AutoML, and run ML directly in SQL with BigQuery ML
- 4 **Implement conversational AI** — build enterprise chatbots and voice agents with Dialogflow CX, including multi-turn flows and webhook integration
- 5 **Process documents at scale** — Google Document AI for invoices, contracts, forms, and custom document types with extraction and classification
- 6 **Deploy and monitor AI** — Vertex AI endpoints, model monitoring, A/B testing, Cloud Functions for serverless inference, and cost optimization

Prerequisites

Python programming (intermediate), basic GCP familiarity (console, projects, IAM). SQL knowledge for BigQuery labs. A GCP account with billing enabled is recommended (free tier credits work for labs).

5-Day Curriculum

Entirely hands-on with GCP console and SDK labs. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Gemini API & Google AI Studio

LIVE CODING + HANDS-ON LABS

SESSION A

Gemini API Fundamentals

- Google AI ecosystem overview: Gemini, Vertex AI, AI Studio
- Gemini models: Flash, Pro, Ultra — capabilities and selection
- Chat completions, system instructions, safety settings
- Lab: Build a Gemini chat application with Google AI SDK
- Multimodal inputs: image, video, audio, PDF analysis

SESSION B

Advanced Gemini Patterns

- Function calling and tool use with Gemini
- Grounding with Google Search and enterprise data
- Structured output, JSON mode, and schema enforcement
- Lab: Build a multimodal product catalog assistant
- Lab: Build a document analyzer that processes PDFs and images

02

Vertex AI Platform & Model Deployment

HANDS-ON LABS + ARCHITECTURE WORKSHOP

SESSION A

Vertex AI Core Platform

- Vertex AI Workbench: notebooks, experiments, datasets
- Model training: custom training jobs, hyperparameter tuning
- Model registry: versioning, staging, deployment
- Lab: Train and deploy a custom model on Vertex AI
- Lab: Run hyperparameter tuning with Vertex AI Vizier

SESSION B

Vertex AI Pipelines & MLOps

- Vertex AI Pipelines with Kubeflow Pipelines SDK
- Pipeline components: training, evaluation, deployment
- Model monitoring: drift detection, feature attribution
- Lab: Build an end-to-end ML pipeline with Vertex AI
- Lab: Set up model monitoring with alerting

03

RAG, Vector Search & Generative AI on GCP

LIVE CODING + HANDS-ON LABS

SESSION A**RAG with Vertex AI Search**

- Vertex AI Search: enterprise search with AI-powered ranking
- Embeddings with Vertex AI text-embedding models
- Vector Search: building and querying vector indexes
- Lab: Build a RAG pipeline with Vertex AI Search + Gemini
- Lab: Enterprise document Q&A with grounded responses

SESSION B**Generative AI Applications on GCP**

- Vertex AI Generative AI Studio: prompt management, tuning
- Model Garden: open-source models (Llama, Mistral) on GCP
- Model fine-tuning: supervised and RLHF approaches
- Lab: Fine-tune Gemini on custom enterprise data
- Lab: Deploy an open-source LLM from Model Garden

04

Dialogflow CX, Document AI & BigQuery ML

HANDS-ON LABS + INDUSTRY PROJECTS

SESSION A**Dialogflow CX & Document AI**

- Dialogflow CX: flows, pages, intents, entities, webhooks
- Building multi-turn conversational agents
- Document AI: processors for invoices, receipts, contracts
- Lab: Build a customer support bot with Dialogflow CX
- Lab: Build an invoice processing pipeline with Document AI

SESSION B**BigQuery ML & AutoML**

- BigQuery ML: ML models in SQL (classification, regression, clustering)
- AutoML Tables: no-code model training for tabular data
- AutoML Vision and AutoML Natural Language
- Lab: Build a churn prediction model using BigQuery ML (SQL only)
- Lab: Train a custom image classifier with AutoML Vision

05

Vision AI, Speech & Capstone

CAPSTONE PROJECT + ARCHITECTURE REVIEW + PRESENTATIONS

SESSION A**Cloud Vision, Speech & Translation**

- Cloud Vision API: image analysis, OCR, label detection
- Cloud Speech-to-Text and Text-to-Speech
- Cloud Translation API: real-time multilingual support
- Cloud Natural Language API: sentiment, entity, syntax analysis
- Lab: Build a multilingual content analysis pipeline

SESSION B**Capstone: Enterprise GCP AI Solution**

- Team project: End-to-end GCP AI solution
- Gemini + Vertex AI + Document AI + BigQuery integration
- Architecture review: IAM, networking, cost, data residency
- Team presentations and expert feedback
- Google Cloud certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest Google Cloud AI developments.

Google Cloud AI Services Covered

All services used hands-on with GCP console and SDK labs

Gemini API

Vertex AI

Vertex AI Search

Vertex AI Pipelines

AutoML

BigQuery ML

Dialogflow CX

Document AI

Cloud Vision API

Cloud Speech

Cloud Translation

Natural Language API

Model Garden

Cloud Functions

Cloud Run

What You Take Home

 **6 GCP AI Projects**

Complete code — Gemini chat app, RAG pipeline, Dialogflow bot, Document AI extractor, BigQuery ML model, Vision pipeline

 **GCP AI Architecture Guide**

Reference patterns for enterprise GCP AI: IAM, VPC-SC, data residency, cost optimization, multi-region deployment

 **Vertex AI Pipeline Templates**

Reusable Kubeflow pipeline components for training, evaluation, deployment, and monitoring on Vertex AI

 **GCP vs Azure vs AWS Matrix**

Detailed comparison of AI services across all three clouds — features, pricing, strengths, migration patterns

Certification of Completion

 **Gennoor Tech — Certificate of Completion**

Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "AI on Google Cloud". Digitally signed by the Founder & Director. Validates proficiency in Gemini API, Vertex AI, Document AI, Dialogflow CX, BigQuery ML, and enterprise GCP AI architecture.

Recommended Google Certifications

Google PMLE
Professional ML Engineer

Google PCDE
Professional Cloud Data Engineer

Google ACE
Associate Cloud Engineer

Google PCA
Professional Cloud Architect

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited

Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of experience delivering AI programs across Azure, GCP, and AWS. Multi-cloud expertise covering Vertex AI, Azure AI, and SageMaker. Delivered AI training for Microsoft, EY, IBM, Boeing, and 80+ organizations across 6+ countries.

12+ active certifications across Microsoft, GitHub, and cloud platforms. Expertise in multimodal AI, RAG pipelines, and conversational AI across all major cloud providers. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

 **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Build AI on Google Cloud?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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